Module 2 Summary

After completing all of the materials, please take a moment to reflect on what was covered. The learning objectives below capture the key takeaways from this module:

- Describe the main assumption of linear regression.
- Write down the assumed equation for the dependent variable.
- Define the family of functions from which $f(X)$ comes from.
- List what shapes $f(X)$ can be.
- Understand why the estimated model is different from the actual model, which explains the “hat” notation.
- Define the loss function used to perform linear regression.
- Describe the mathematical process of constructing a linear model.
- List all the assumptions required for linear regression.
- Define the residual.
- State the distribution of $\hat{\beta}$.
- State the estimate for the variance of the residuals, denoted $\hat{\sigma}^2$.
- Calculate a t-score to test if a given $\beta_j = 0$.
- State how to construct an F-statistic to test the significance of a group of coefficients.
- Know the results of the Gauss-Markov Theorem for ordinary least squares regression.
- Describe subset selection and its three main techniques.
- List strengths and weaknesses of the technique.
- List two methods for model evaluation with subset selection.
- Define data snooping.
- Describe regularization and define hyperparameter.
- Give the formula for Ridge Regression and explain how it works.
- Describe the benefits and drawbacks of ridge regression.
- Define LASSO regression.
- Explain why LASSO is different from Ridge.
- Describe the strengths and weaknesses of LASSO.
- Explain why data standardization can be important and how to do it.
- List types of data transformations commonly done with linear regression and explain why they are necessary.
- List examples where a data transformation is warranted.